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Review

A Systematic Review of Accessibility Techniques for Online Platforms: Current Trends and Challenges

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Abstract: Accessibility in online platforms is a critical concern in our increasingly digital world, where information and services are predominantly accessed through the Internet. The purpose of this systematic review is to provide a comprehensive overview of the current state of the art in online accessibility technologies, and it is focused on key tools such as sign language recognition, speech-to-text, text-to-speech, and voice recognition. Despite advancements in digital inclusivity, numerous technical limitations persist, which limit the accessibility of online content for individuals with disabilities. Our findings indicate that while speech and voice technologies have achieved good accuracies and low word error rates, further research is needed to improve the accuracy and usability of sign language recognition systems, especially for continuous sign language recognition, as they have low accuracy. In this review, we analyzed research articles and publications from well-known databases, including Google Scholar, Elsevier, IEEE Xplore, and Springer. In order to ensure a high standard of quality, we applied the PRISMA 2020 and PEDro methodologies to quantitatively and qualitatively filter the thousands of articles provided by these databases, and we selected only studies that were related to our study. Key areas of investigation included the performance and accuracy of sign language interfaces, speech-to-text, text-to-speech, and speech recognition applications and the compatibility of these technologies with different platforms and devices. This review also explores the role of emerging technologies such as artificial intelligence (AI) and machine learning (ML) in enhancing accessibility and personalizing user experiences. Through a critical analysis of current solutions and a discussion of existing gaps, this paper offers insights into potential improvements and future directions for creating more accessible online environments. The findings might be valuable to researchers and developers dedicated to promoting digital inclusivity and equality.

Keywords: sign language recognition; accessibility and interfaces; machine learning; artificial intelligence; text-to-speech; speech-to-text; voice recognition

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1. Introduction

Currently, there are 1.3 billion people with different types of disabilities [1] who face significant challenges when it comes to using online platforms. From an accessibility point of view, many of these individuals encounter a multitude of barriers that limit their ability to access and interact with web content effectively. In some cases, the lack of proper accessibility features can make certain platforms completely unusable for individuals with specific needs, so they are excluded from the benefits and opportunities offered by the digital world. This situation not only impacts the personal and social lives of those affected but also limits their ability to participate fully in education, employment, and other essential activities.

In order to address these issues, there are internationally recognized standards and guidelines that web developers and designers are expected to follow when creating online platforms. One of the most recognized and largely adopted sets of guidelines is known as

the Web Content Accessibility Guidelines (WCAG) 2.1. These guidelines are developed by the World Wide Web Consortium, and they provide a comprehensive framework aimed at making web content more accessible to a broader range of users, including individuals with disabilities. The WCAG 2.1 revolve around various principles and criteria designed to accommodate people with diverse needs such as blindness and low vision, deafness and hearing loss, limited mobility, speech disabilities, cognitive and neurological impairments, and others. By adhering to these guidelines, developers can create websites that are perceivable, operable, understandable, and robust, thus promoting a more inclusive digital environment [2].

Despite the existence of these standards, implementing effective accessibility solutions often requires the integration of advanced technologies such as ML algorithms. These algorithms play a crucial role in developing interfaces designed for specific disabilities by providing functionalities like text-to-speech (TTS) and speech recognition (SR) for those with visual impairments, speech-to-text (STT) for individuals with hearing loss, and sign language recognition (SLR) for those with speech or auditory disabilities. Additionally, other assistive technologies may include alternative navigation methods for those with limited mobility, predictive text input, and automated captioning for video content. The application of ML and AI in these areas has the potential to significantly enhance user experience (UX) by creating more intuitive, responsive, and personalized interactions for users with diverse accessibility needs.

While there has been considerable progress in accessibility tools like STT, TTS, and voice recognition, several critical gaps remain, particularly in the area of SLR. Sign language recognition still lacks the precision necessary for consistent usability, especially in continuous video contexts [3]. As a result, accessibility solutions remain inconsistent, with many platforms still unable to accommodate users with specific needs fully.

Through this systematic review, we aim to thoroughly explore and analyze the current state of the art in developing accessible online interfaces that incorporate ML algorithms, particularly analyzing the accuracy, precision, and word error rates of these algorithms. By examining the latest research and advancements in the field, we seek to identify gaps and limitations in existing technologies and methodologies. Our goal is to highlight areas where improvements are necessary to achieve better algorithm accuracy, enhance UX, or make previously inaccessible platforms more usable.

This review will provide valuable insights for researchers, developers, and stakeholders involved in the creation of inclusive digital environments, guiding future efforts to optimize accessibility and promote digital equity.

Our research paper aims to answer the following questions:

- What are the similarity scores for speech-to-text algorithms and word-level intelligibility of text-to-speech algorithms?
- What are the word error rates and recognition rates of speech recognition algorithms?
- Are there sign language recognition algorithms that have an accuracy higher than 95% for continuous SLR?
- What are the methods used to build sign language recognition algorithms?

In the next chapter, we will describe the methodology we used to filter the selected articles that contain valuable information such as accuracy/precision of ML algorithms that could be used to implement accessible interfaces (STT, TTS, SR, and SLR) for people suffering from speech impairments, deafness, and blindness.

2. Research Methodology

This section outlines the protocol used to identify, gather, and evaluate the state-of-the-art techniques examined in this systematic review. The process is divided into four key phases: development of the research strategy, formulation of the research questions, criteria for article selection, and presentation of the next steps in our research.

2.1. Developing the Research Strategy

In our study, we put together a solid collection of research papers on our topic to obtain a well-rounded view and an in-depth understanding of the latest progress and advancements in accessibility technologies. The articles we collected were published between 2018 and 2024, representing the most recent developments in this field. To ensure a wider scope, we used papers from multiple reputable databases such as Google Scholar, IEEE Xplore, Elsevier, and Springer. Our search strategy involved using various combinations of keywords and key phrases, such as "Sign Language Recognition", "Developing Accessible Interfaces", "Speech Recognition", "Speech to Text Interfaces", and "Text to Speech Interfaces". This extensive search initially returned 2819 articles.

We applied the PRISMA and PEDro methodologies to rigorously filter the large volume of studies identified in our initial search. PRISMA provided a structured approach for organizing and documenting the selection process. By detailing each step of the study selection process, PRISMA allowed us to systematically narrow down the results to the studies that were most relevant to our focus on online accessibility technologies. Meanwhile, the PEDro scale was essential for assessing the methodological quality of these studies.

Given the large number of articles received, we implemented a multi-stage research strategy that involved both quantitative and qualitative filtering to identify the most relevant and high-quality studies. One of the first challenges we encountered in our filtering process was the large number of duplicate articles. In order to solve this, we had to aggregate all the results from the searches through an Excel file and use Microsoft Excel features to identify the duplicate results and remove them. After this step, the number of articles was reduced by 446 articles, leaving us with 2373 unique papers.

Following this, we applied a quantitative filter based on specific paper characteristics. We required each article to have been cited at least ten times, indicating a certain level of recognition and impact within the community. After applying these criteria, we reduced the pool to 2285 articles.

We conducted a more targeted filtering process by reviewing the titles and abstracts of the remaining articles. During this phase, we filtered the title and abstract with additional keywords, including "Accessible Interface", "Speech to text", "Text to speech", "accessibility", "Speech recognition", and "sign language recognition". This step allowed us to exclude 2176 articles that did not align closely with our main research focus. After this extensive screening, 109 articles appeared directly relevant to our study.

From these 109 articles, we applied a qualitative filtering based on a set of questions that will be described in the following subsection. After this step, 35 articles remained. Figure 1 [4] shows how the research strategy steps were applied to the results. On the other hand, in Table 1 are the results from the research databases based on the queries used for the search.

Table 1. Number of results returned by queries from each database.

Query	Elsevier	IEEE	Springer	Google Scholar	Total
Developing accessible interfaces	49	37	80	332	497
Text to speech interfaces	46	174	122	151	493
Sign language recognition	137	124	123	436	820
Speech to text interfaces	37	176	120	158	491
Speech recognition	142	149	91	136	518
Total	411	659	536	1213	2819

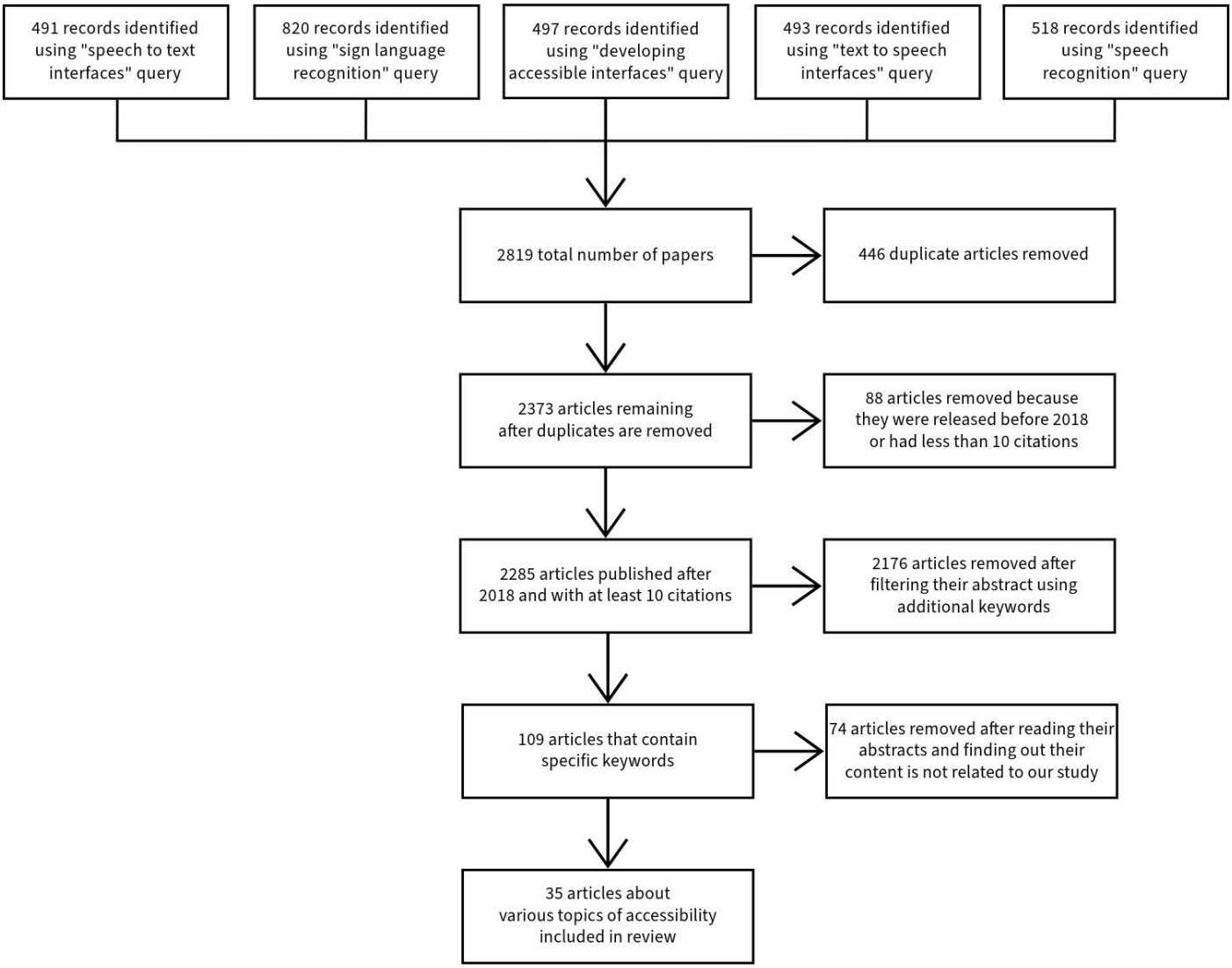


Figure 1. PRISMA diagram flow of selecting the articles (PEDro filters included).

2.2. Formulation of the Research Questions

After completing the quantitative filtering using PRISMA filtering, we conducted qualitative filtering using the PEDro methodology to obtain more relevant results related to our main research topic. The qualitative filtering involved asking a set of questions that each article should address. If a research paper answered at least two questions from the filtering process, we included it. The questions for qualitative filtering were the following:

1. What are the methods used to implement an accessible interface?
2. Is it using voice recognition, text-to-speech, speech-to-text, or sign language recognition algorithms?
3. Is the research aiming to help the vulnerable categories of people with disabilities?
4. Is the machine learning algorithm producing good results and does it have good accuracy?
5. Is the proposed solution implemented, adapted, or proposed to a real-life use case?

As we have mentioned before, after the qualitative filtering, we removed 74 articles from the 109 initially selected, leaving 35 remaining articles.

2.3. Criteria for Article Selection

As a conclusion based on what we said in the previous subsections, the criteria for article selection were the following:

1. Quantitative filtering (PRISMA filtering):
   (a) Articles published between 2018–2024;
   (b) Articles that were written only in the English language;
   (c) Articles that had at least 10 citations;
   (d) Articles that had some keywords related to our study in the title or abstract.
2. Qualitative filtering (PEDro filtering):
   (a) Articles that implemented accessible interfaces;
   (b) Articles that implemented or used machine learning algorithms that could be used to implement accessible interfaces;
   (c) Articles that aimed to help people with disabilities;
   (d) Articles that had results (either bad or good) and their results.

It is important to mention that firstly, we applied quantitative filtering to narrow down the pool of articles, and then qualitative filtering was applied. In the qualitative filtering, we read the abstract of the 109 remaining papers, and we searched for specific topics such as speech-to-text, speech recognition, text-to-speech, and sign language recognition. We removed the articles that were not aligned or related to our study (e.g., accessibility not related to machine learning or artificial intelligence).

2.4. Next Steps in Our Research

After conducting searches across the scientific databases and identifying relevant research, we selected articles that specifically explored various techniques for developing accessible interfaces for individuals with disabilities. Each of these articles was reviewed, and the findings were analyzed to summarize the current state of research, identify the most effective approaches, and highlight potential areas for future exploration.

3. Results of the Systematic Review

In this section, we will provide a comprehensive discussion of all the articles selected from the previous section. We will go through the key findings and insights from each study, offering a detailed analysis of the most significant results.

In the first three subsections, we attached some charts representing the trending of accessibility keywords, such as “speech recognition”, “sign language recognition” etc. Numbers represent search interest relative to the highest point on the chart for the given region and time. A value of 100 is the peak popularity for the term. A value of 50 means that the term is half as popular. A score of 0 means there were not enough data for this term [5].

3.1. Relevant Solutions Based on Speech Recognition

Voice recognition technology powers secure user authentication and personalized experiences. Applications include voice-based biometrics in banking (e.g., Wells Fargo’s voice authentication [6]), smart home devices (e.g., Google Nest), and call centers for customer identity verification. Challenges in voice recognition include handling variations in voice due to health, age, and emotional state, which can lead to inaccuracies and frustrations in sensitive contexts like financial services.

Advancements in ASR technologies have shown varied accuracy rates among leading solutions, with notable performances from providers like Whisper and Speechmatics [7]. These findings highlight the importance of evaluating ASR systems not only for their accuracy but also for their accessibility features. Research indicates that voice-controlled personal assistants, such as the Amazon Echo, are beneficial for users with disabilities, particularly for those with speech impairments [8]. The positive user feedback underscores the potential of these technologies to enhance communication and interaction in daily life.

Subhash, S. et al. discuss in their work the growing role of voice control technology, particularly AI-based voice assistants used in smartphones and laptops [9]. These

assistants recognize human voice, convert it to text, and then use the Google text-to-speech (GTTS) engine to produce audio in English, which is played back using Python's (both 2.7 and 3.x versions) play sound package. Also, the study of Pradhan, A. et al. explores the accessibility of voice-controlled personal assistants like the Amazon Echo for users with disabilities. Analysis of 346 reviews and interviews with 16 visually impaired users show that, despite some challenges, many, including those with speech impairments, find the device helpful, with 59.4% of reviews on SR being positive and 74.2% of speech-impaired users reporting satisfaction [8].

Another interesting study conducted by Branham, S. M. et al. says that voice-activated personal assistants (VAPAs), like Siri and Alexa, benefit users, especially those with visual impairments [10]. This study reviews design guidelines from major companies, highlighting the assumption that VAPAs should mimic human conversation. It critiques this approach and suggests improvements to enhance usability for all users. On the other hand, Blue, L. et al. demonstrates that the Two Microphone Authentication system can accurately localize audio sources within a narrow cone of less than 30° while achieving zero false positives [11]. It effectively eliminates replay attacks and prevents the injection of inaudible or hidden commands, significantly enhancing the security of voice-controlled interfaces. Canonico, M. et al. review and compare various cloud-based natural language understanding platforms developed over the last decade. It presents a taxonomy for six platforms and conducts a performance evaluation of three of them, highlighting their respective strengths and weaknesses. The analysis reflects the widespread adoption of these tools in chat services and conversational assistants on mobile devices [12]. Darabkh, K. A. et al. present an SR system designed to help arm-disabled students to control computers by voice, enhancing accessibility in education. The system uses two approaches: one based on voice activity detection and Mel-frequency cepstral coefficients and another using discrete wavelet transform with modified MFCC. The first approach achieved a recognition rate of 98.7%, while the second achieved a rate of 98.86%, offering higher accuracy but slower processing [13]. In order to explore recent advancements in accessibility research, Brewer, R. N. et al. situates voice-based accessibility within existing theoretical frameworks, discusses design challenges, and identifies opportunities for interdisciplinary collaboration in this area [14]. In the same context, Kuhn, K. et al. highlight the measuring of accuracy of ASR solutions. The word error rates were as follows: 22.4% for Google, 12.4% for IBM, 10.8% for Deepgram, 10.5% for SpeechText.AI, 5.6% for Rev AI, 5.3% for Amazon and Microsoft, 4.7% for Tencent, 4.3% for AssemblyAI, 3.4% for Speechmatics, and 2.9% for Whisper [7].

The advancements in SR algorithms have reached a mature level, demonstrating significant improvements in accuracy and functionality across various applications. Leading solutions have achieved impressively low word error rates, indicating a high degree of reliability in real-world usage.

In Figures 2 and 3 are the trends of the keywords "voice recognition" and "speech recognition" over the last 15 years.

From these two figures, we can conclude that voice and speech recognition technologies have matured and become widely integrated into a multitude of devices (e.g., smartphones, smart speakers, cars, etc.), such that people may view them as a standard feature rather than a novel technology. This normalization can lead to fewer searches since consumers are no longer actively learning about or exploring the technology; they simply expect it to be there.

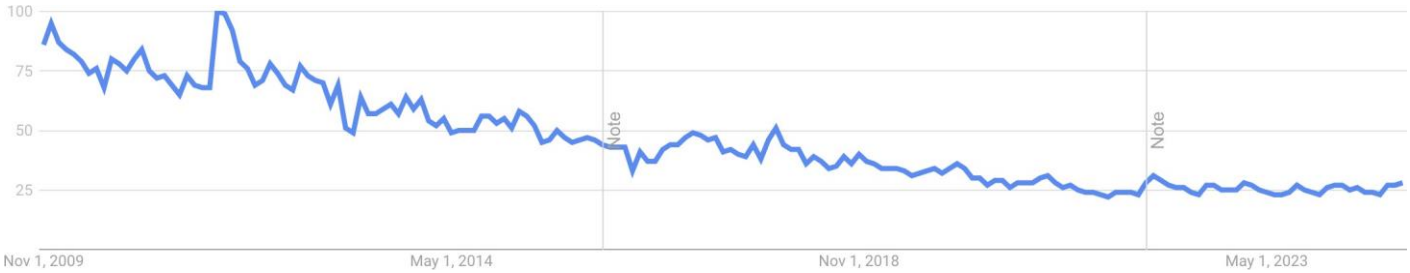


Figure 2. The trending of the keyword "voice recognition" over the last 15 years (November 2009–November 2024) from Google Trends.

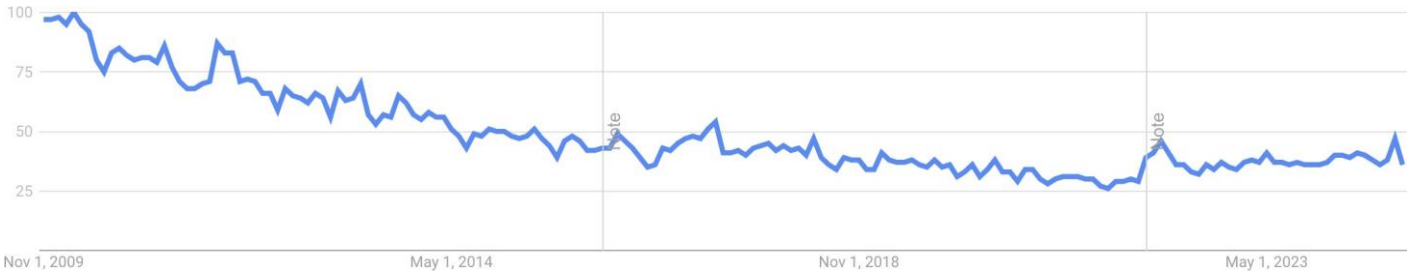


Figure 3. The trending of the keyword "speech recognition" over the last 15 years (November 2009–November 2024) from Google Trends.

3.2. Solutions Based on Speech-to-Text and Text-to-Speech

Speech-to-text technology has seen significant advancements in accuracy, largely due to improvements in natural language processing and deep learning models. It is widely used in applications like transcription services (e.g., Otter.ai and Google Live Transcribe), virtual assistants (e.g., Siri, Alexa), and closed captioning for accessibility in media. Real-world challenges include handling diverse accents, noisy environments, and domain-specific terminology, which can affect user satisfaction and the overall effectiveness of STT.

Text-to-speech systems have advanced to generate more natural and expressive voices. These systems are now integral to assistive technologies like screen readers for visually impaired users (e.g., JAWS and NVDA) and language learning apps (e.g., Duolingo). TTS are integrated in many real-world applications such as GPS systems that provide spoken directions, virtual assistants (e.g., Siri, Google Assistant, Alexa), and accessibility tools for individuals with visual impairments (e.g., screen readers like JAWS and VoiceOver).

Recent advancements in TTS technology have led to remarkable accuracy and performance levels, particularly in models like Tacotron 2, which achieved an MOS of 4.38, and Tacotron 1, which produced an MOS of 4.01 [15]. Additionally, an almost unsupervised learning method for TTS aimed at low-resource languages demonstrated exceptional results, with a 99.84% word-level intelligibility but a 2.68 MOS on the LJSpeech dataset, despite utilizing only 200 paired data points and additional unpaired data [16].

Zen, H. et al. present "LibriTTS", a speech corpus designed for TTS applications, based on the LibriSpeech dataset. It includes 585 h of 24 kHz speech data from 2456 speakers. Models trained on LibriTTS achieved mean opinion scores (MOS) above 4.0 for naturalness in five out of six evaluation speakers [17]. Trivedi, A. et al. explore applications that facilitate text and speech communication, including SR, STT conversion, synthetic speech generation, and language translation. The focus is on the methods employed to ensure seamless interaction between parties across networks [18]. In the same context, Khan, A. et al. introduce a blind-friendly user interface framework that simplifies interface components based on user profiles and context. An empirical study with 41 participants demonstrated significant improvements in their smartphone interaction experience [19].

Anandan, M. et al. present an object detection and navigation system for visually impaired individuals, featuring static and dynamic obstacle detection. The static module uses techniques like SURF feature extraction, while the dynamic module employs

Gaussian mixture models [20]. Visual information is converted into audio feedback via a Raspberry Pi, including vibrations for users with partial hearing loss. The system effectively models backgrounds, eliminates shadows, and supports blind users in navigating their environment.

Aguilar-Chacon, J. E. et al. explore the integration of small-sized computers with STT technology, particularly addressing the challenge of implementing STT algorithms on microcontroller-based systems without compromising performance. A methodology is presented to evaluate STT services using similarity and speed as metrics, with an experiment conducted on a Raspberry Pi 3 for Spanish speakers. The results reveal that Google's STT system outperformed both IBM and Microsoft, showing faster conversion times and higher accuracy, with a similarity score of 97.116%, compared to Microsoft's 90.825% and IBM's 90.337% [21].

Following the experiments of other researchers in this field, Khanam, F. et al. created a literature review where they discuss the process of TTS synthesis, which converts natural-language text into speech using recorded samples stored in a database. The review highlights the significant advancements in TTS due to deep learning, compared to traditional machine learning methods. The paper introduces a taxonomy of popular deep learning architectures and models used in speech synthesis, reviews various datasets utilized in TTS, and describes common evaluation metrics for assessing synthesized speech quality [15]. It concludes by outlining the challenges and future directions for TTS systems. Ren, Y. et al. present an almost unsupervised learning method for TTS and automatic speech recognition (ASR) aimed at low-resource languages, addressing the challenge of limited aligned data. The approach combines several components: a denoising auto-encoder for language modeling; dual transformation to enhance accuracy between TTS and ASR; bidirectional sequence modeling to reduce error propagation; and a unified transformer-based model structure. The method achieves 99.84% word-level intelligibility, a 2.68 MOS for TTS, and an 11.7% phoneme error rate for ASR on the LJSpeech dataset, using only 200 paired data points (about 20 min of audio) and additional unpaired data [16].

The trend lines shown in Figures 4 and 5 for the keywords "speech to text" and "text to speech" appear quite similar over the past 15 years. Both terms show relatively low and stable interest in the earlier years, followed by an increase in interest around 2021. This similarity suggests a correlated rise in the relevance of both technologies, possibly due to the increasing adoption of voice-enabled applications, digital assistants, and accessibility solutions.

The simultaneous increase in search interest might indicate that users are increasingly seeking or using both "speech to text" and "text to speech" functionalities, potentially driven by advancements in AI, improvements in accessibility technologies, and the rise of remote work and online communication. This parallel growth could reflect a broader trend toward integrating voice-based technologies in online platforms and other digital environments.

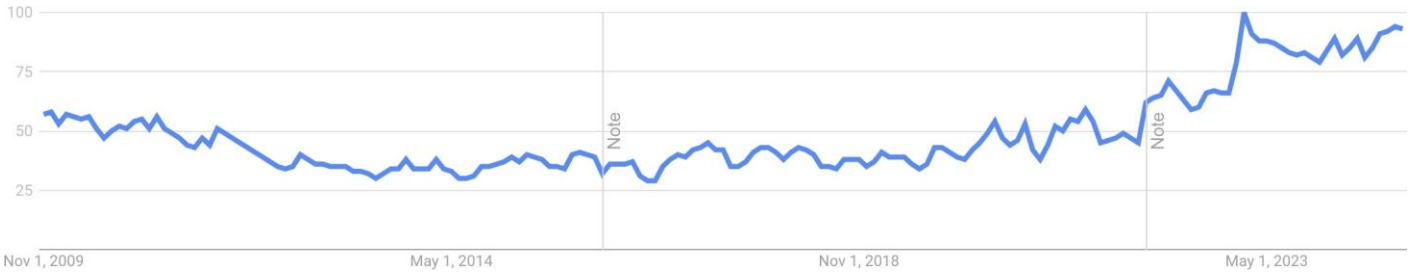


Figure 4. The trending of the keywords "speech to text" over the last 15 years (November 2009–November 2024) from Google Trends.

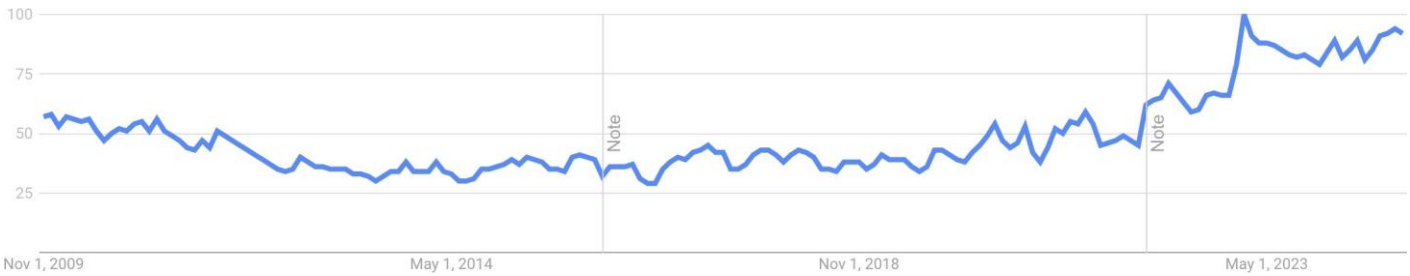


Figure 5. The trending of the keywords "text to speech" over the last 15 years (November 2009–November 2024) from Google Trends.

3.3. Advanced Solutions on Sign Language Recognition

SLR is a newer and less mature field compared to other recognition technologies, often limited by expensive datasets and language variations [22]. Current applications are mainly in experimental settings, but some real-world implementations are emerging, like translating sign language for customer service desks (e.g., SignAll's software). Challenges include capturing nuanced hand and facial expressions, high computational requirements, and adapting to different sign languages, which limit SLR's broader adoption and user satisfaction.

Recent advancements in SLR have demonstrated impressive accuracy levels across various models and approaches. For instance, a model trained on 54,049 images of Arabic sign language achieved a precision rate of 95% [23]. In another study, the SAM-SLR framework integrated multiple modalities and reported recognition rates of 98.42% for RGB images and 98.53% for RGBD [24]. Additionally, a hierarchical model for ISL gestures reached 98.52% accuracy for one-hand gestures [25], while a wearable SLR device using electromyography and inertial sensors achieved an average accuracy of 96.13% [26]. These achievements highlight the potential of advanced techniques in enhancing the reliability of SLR systems.

Zakariah, M. et al. trained a model using 54,049 images with Arabic sign language. The model had a precision rate of 95% [23]. Also, Rastgoo, R. et al. conducted a survey that reviews recent vision-based sign language recognition models utilizing deep learning, highlighting accuracy improvements and ongoing challenges. It categorizes models for isolated and continuous recognition, discussing applications, datasets, hybrid approaches, and future research directions [27]. Jiang, S. et al. propose a novel multi-modal framework, SAM-SLR, for SLR that integrates skeleton-based, RGB, and depth modalities to enhance recognition accuracy. The model achieved a good performance, with recognition rates of 98.42% for RGB and 98.53% for RGBD, making it the top performer in the 2021 SLR Challenge [24].

Wen, F. et al. developed an AI-enabled sign language recognition system combining sensing gloves, deep learning, and a virtual reality interface. Their model recognized 50 words and 20 sentences, achieving an 86.67% accuracy for new sentence constructions by recombining word elements [28]. Camgoz, N. C. et al. present a transformer-based model that jointly learns sign language recognition and translation, achieving state-of-the-art results on the PHOENIX14T dataset. Their approach significantly improves translation performance, with BLEU-4 scores rising from 9.58 to 21.80, and sets new baselines for sign language translation tasks [29]. Wadhawan, A. et al. present a deep-learning-based convolutional neural network (CNN) model for Indian sign language (ISL) recognition, using 35,000 images of 100 static signs. The system, evaluated on 50 CNN models, achieves a top training accuracy of 99.72% for colored images and 99.90% for grayscale images [30]. Li, D. et al. introduce the large-scale Word-Level American Sign Language (ASL) dataset, featuring over 2000 words performed by more than 100 signers, the largest public ASL dataset available. The authors evaluate two deep learning models for word-level sign recognition: a holistic visual appearance-based approach and a 2D human pose-based approach, achieving a top-10 result of up to 62.63% accuracy [31].

A robust analysis of gesture recognition techniques was performed by Sharma, A. et al. for ISL using three models: fine-tuned VGG16, transfer learning VGG16, and a hierarchical neural network. The hierarchical model achieved the best accuracy, with 98.52% for one-hand gestures and 97% for two-hand gestures [25]. Khomami, S. A. et al. present a wearable low-cost SLR device using surface electromyography and inertial measurement unit sensors to enhance accuracy. Data from ten volunteers performing 20 Persian sign language signs were collected, with 25 key features extracted and classified using a KNN classifier, achieving an average accuracy of 96.13% [26]. In the same context, Kunjumon, J. et al. propose a system that recognizes ISL and converts it to speech and text in English and Malayalam, displaying the output on an Android device, with an accuracy of approximately 90% [32].

Rao, G. A. et al. address the challenge of recognizing ISL gestures using convolutional neural networks. A dataset was created featuring five subjects performing 200 signs from different angles and backgrounds, with each sign captured in 60 frames. The study involved training CNNs with various sample sizes and architectures, achieving a recognition rate of 92.88%, outperforming other classifiers on the same dataset. The proposed method enables independent use of a sign language recognition mobile application by hearing impaired individuals [33]. Huang, J. et al. propose LS-HAN, a framework for continuous sign language recognition that eliminates temporal segmentation. It uses a two-stream CNN and hierarchical attention network, achieving up to 82.7% accuracy [34].

Pu, J. et al. introduce an alignment network with iterative optimization for weakly supervised continuous sign language recognition. It combines a 3D ResNet for feature learning and an encoder–decoder network with CTC for sequence modeling, optimized iteratively. Evaluated on large datasets, the method achieved a precision of 93.9% [35]. Mittal, A. et al. propose a modified LSTM model for continuous SLR, capable of recognizing sequences of connected gestures by splitting them into sub-units. Tested on 942 ISL sentences using 35 sign words, the system achieved an average accuracy of 72.3% for signed sentences and 89.5% for isolated words [3]. Bragg, D. et al. emphasize the need for interdisciplinary collaboration to develop sign language recognition, generation, and translation systems. It summarizes a workshop with experts from various fields, reviewing the state of the art, identifying key challenges, and calling for more integrated research efforts in the community [22]. Murali, R. S. L. et al. introduce a system for recognizing ASL gestures using computer vision techniques. It captures hand gestures via a webcam, processes the images using the HSV color algorithm, and uses a CNN for classification. The model successfully recognizes 10 ASL alphabets with above 90% accuracy [36]. Kasapbaşi, A. et al. developed an ASL recognition dataset and a CNN model to interpret ASL gestures into natural language. The dataset accounts for various conditions, marking a significant advancement in sign language recognition. The CNN achieved 99.38% accuracy with a low loss of 0.0250, outperforming two existing datasets. These results suggest that the system could be a promising solution for medical applications using deep learning while enhancing contributions to the SLR field [37].

Bantupalli, K. et al. created a vision-based application that translates sign language into text, enhancing communication between signers and non-signers. It processes video sequences to extract features, using an inception CNN for spatial recognition and an RNN for temporal training, based on the ASL dataset. The accuracy using a Softmax layer was around 90% [38].

However, continuous sign language recognition from videos presents more significant challenges, often resulting in lower accuracy rates. While static recognition benefits from clear and defined gestures, dynamic recognition must contend with variations in speed, context, and environmental factors, which can limit performance. For instance, despite advancements, the overall effectiveness of recognizing continuous signing from videos has not reached the same levels of accuracy seen with individual images. Addressing these challenges will be crucial for improving real-time, continuous sign language recognition systems.

We can observe in Figure 6 the trending of the keywords "sign language recognition" over the last 15 years and can conclude that the chart indicates that interest in sign language recognition was minimal or nearly non-existent from 2009 until the end of 2021. This trend changed noticeably in the last quarter of 2021 and the first quarter of 2022, where there was a sudden increase in searches or interest.

This trend suggests that while sign language recognition was a niche topic for over a decade, recent developments—technological, social, or research-based—have brought it into broader focus. This emerging interest may indicate both a recognition of the need for such technologies and a signal of future growth in the field.

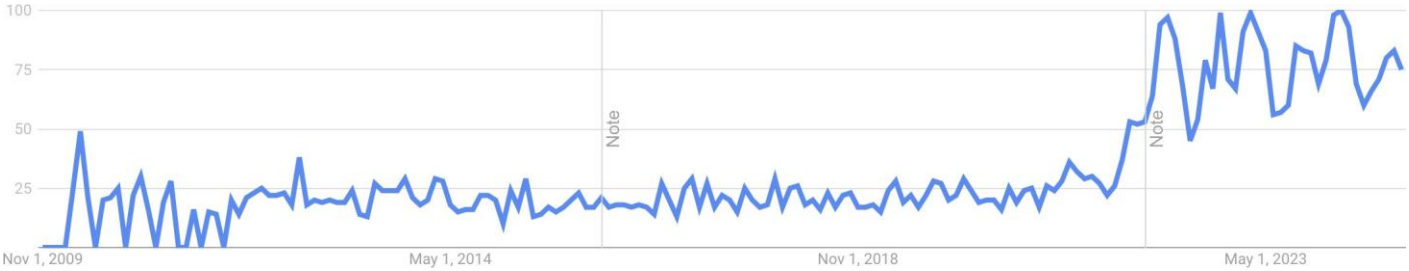


Figure 6. The trending of the keywords "sign language recognition" over the last 15 years (November 2009–November 2024) from Google Trends.

3.4. Other Relevant Studies

There were also articles that highlighted the potential for AI technologies to significantly impact people with disabilities (PWD), especially through tools like automated SR and language prediction systems. While these AI systems can enhance accessibility, they can also risk marginalizing PWD if not designed and tested carefully. The paper calls attention to fairness in AI for PWD, identifying concerns about how AI systems may fail or discriminate if care is not taken in development [39].

In other articles, authors discuss the impact of artificial intelligence, particularly machine learning, on fair treatment for people with disabilities. This highlights that fairness for this group differs from fairness concerning groups with other protected attributes like age, gender, or race, primarily due to the vast diversity in how disabilities manifest and the sensitive nature of disability information, which is often not shared to avoid discrimination [40].

4. Exploring Future Directions

As we explore the development of accessible interfaces for individuals with disabilities, there are several promising directions for future research. These directions aim to improve the accuracy, functionality, and user experience of sign language recognition systems by leveraging advanced AI technologies and ML. Below, we outline four key research directions that could significantly enhance the field of accessibility and create more inclusive digital environments.

4.1. Training a Sign Language Recognition Model with AI-Generated Images

One potential area to explore is the use of AI-generated images to train SLR models. Training these models requires large datasets of sign language videos or images (since these models use supervised learning), which can be hard and time-consuming to obtain. By using images generated with artificial intelligence, scientists can create datasets that mimic real sign language movements. This approach has the potential to expand existing training data and allow models to learn a wider range of gestures and types of sign language.

In addition, this method can solve the problems related to the lack of sign language datasets for less commonly used sign languages. AI-generated data can provide a scalable solution for creating models that are more inclusive and adaptive. Future research should

focus on evaluating the effectiveness of AI-generated images to improve model performance and optimizing the token engineering used to generate these images to ensure they accurately reflect real gestures.

4.2. Enhancing Model Output Using ChatGPT for Rephrasing

Another interesting direction for future research is to integrate natural language processing tools like ChatGPT to improve the results of sign language recognition models. Although these models can accurately identify and translate signs, the raw output can sometimes be unclear or sound strange in natural language. By using large generative language models to reformulate the results, we could improve the fluency and readability of the recognized text, making the output more intuitive and conversational.

This approach has the potential to improve the overall UX, especially for people who rely on this type of communication system. Users interact more seamlessly with this technology, making it easier for them to communicate more effectively. Further research could investigate the impact of this reformulation on user satisfaction and the overall quality of communication in different contexts, such as educational, professional, or social.

In Figure 7, we present the pipeline of how sign language is transposed to plain text and, using a large language model, how the result is rephrased to sound better.

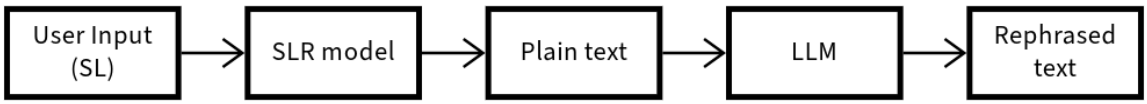


Figure 7. Enhancing SLR model using LLM.

4.3. Sign Language to Enhanced Text (Using LLM) to Speech

Another promising application refers to the conversion of the result of the sign language recognition model into enhanced text using large generative language models, followed by TTS. This development could have a transformative impact on people with speech and hearing impairments who need to deliver speeches/presentations or participate in public speaking. By translating sign language into clear and structured spoken language, TTS systems can help users effectively communicate their messages to a wider audience without relying on intermediaries such as translators.

In addition, integrating tools such as ChatGPT to rephrase the result can ensure that the generated speech is not only accurate but also designed to sound natural and engaging. This would allow users to retain full control over their communication and support greater independence in professional, educational, and social settings.

Figure 8 presents a follow-up scenario, in which sign language is transposed to plain text and, using a large language model, the result is rephrased to sound better, and a text-to-speech model is then used to help the presenter to communicate effectively.

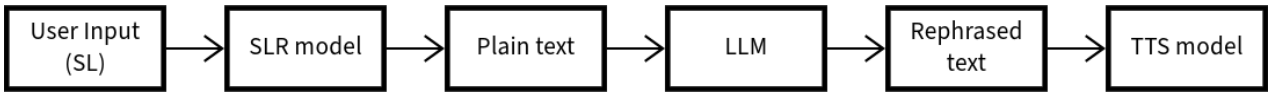


Figure 8. Using the enhanced rephrased text in a text-to-speech model.

4.4. Translating Sign Language Model Output into Multiple Languages

In today's globalized world, language barriers present significant challenges for accessibility. To address this problem, we extend the use of ChatGPT to translate the output of sign language recognition models based on the available languages. This allows the system to not only recognize and annotate sign language to text but also provide translation when other languages are used. This feature improves the accessibility of online platforms for users who are hearing-impaired and who are non-native speakers of the primary language.

This multilingual ability opens the door to wider use, making digital content more accessible. Researchers could focus on assessing the accuracy of these translations, especially in relation to context-specific language usage, idiomatic expressions, and cultural nuances. Another important consideration is how process analysis impacts the overall UX and how it can be optimized for real-time interactions.

Figure 9 presents another use case, in which sign language is transposed to plain text using an SLR model, and the result is translated into other languages using a large language model.

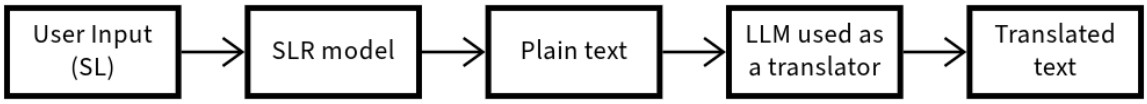


Figure 9. Using LLM to translate the output from SLR model.

4.5. Real-Time Sign Language Interpretation in Livestreams

An innovative direction for future research is the use of sign language recognition models for real-time interpretation during live streams. In this scenario, sign language can be interpreted on the fly and provide instant translation during live broadcasts, conferences, or events. This significantly increases accessibility for people with hearing impairments who participate in this type of real-time interaction.

In this context, ChatGPT can be used to improve interpretation by rewriting the translated content in real-time, ensuring that the result is not only accurate but also contextually appropriate and clear to the audience. This can fill the communication gaps during conversations from events where the original output may be too literal or unclear. Future research could explore the technical challenges of implementing this system in real-time settings, focusing on latency, accuracy, and user feedback to ensure a smooth and effective experience for end users.

5. Conclusions

In order to write this systematic review, we conducted an extensive search across several key academic databases, including Google Scholar, Elsevier, and Springer. This initial search returned thousands of research papers that needed a thorough filtering process. To refine these results and make sure they were closely aligned with the focus of our study, we used the PEDro and PRISMA methodologies. These frameworks allowed us to systematically narrow down the pool of research to obtain more relevant and targeted studies that directly contribute to the objectives of our review.

There are numerous studies that have been conducted with a focus on the development of accessibility solutions through the use of artificial intelligence and machine learning technologies, especially in the areas of speech recognition, sign language recognition, and speech-to-text and text-to-speech conversion. These research efforts highlight the growing interest in the field and the accessibility advancements for individuals with disabilities.

The primary and most significant findings from this study can be summarized as follows:

- Speech-to-text and text-to-speech models have impressive accuracy (e.g., in the Ren, Y. article, they achieved a 99.84% accuracy of word-level intelligibility for an almost unsupervised algorithm of TTS machine learning [16]; the STT model built by Google demonstrated a similarity score of 97.116% after a robust evaluation [21]);
- Whisper's speech recognition algorithm outperformed other models with a word error rate of 2.9% [7]; the algorithm implemented by Darabkh, K. for speech recognition using a discrete wavelet transform with a modified MFCC achieved a remarkable recognition rate of 98.86%;
- Continuous sign language recognition needs improvement because the algorithms implemented by authors in the explored solutions did not achieve more than a 95%

accuracy (Pu, J. achieved a 93.9% accuracy [35], and Mittal, A. achieved a 72.3% accuracy for signed sentences and a 89.5% accuracy for isolated words [3]);
- Research interest in SLR models has increased significantly in the last two years;

In conclusion, while technologies such as STT, TTS, and SR have achieved an impressive level of maturity, this progress is largely attributed to the high accuracy rates (less than 3.5% word error rates for Whisper and Speechmatics [7]) of these systems and their ability to offer outputs with good mean opinion scores (more than 4.38 mean opinion score for Tacotron 2 [15]), which serves as a key measure of user satisfaction. These advancements have made STT, TTS, and SR solutions highly effective and reliable for a wide range of users, including those with speech and hearing impairments. The accuracy of sign language recognition systems remains a critical area where enhancements could lead to better usability and effectiveness for users.

In order to address this challenge, exploring innovative approaches such as AI-generated images for training SLR models could result in insightful and promising results. This method could provide a more diverse dataset, helping, in this way, to capture different variations in sign language. Furthermore, integrating advanced tools like generative large language models such as ChatGPT to rephrase and clarify the output from sign language recognition systems could be a game-changer. Such integration would not only enhance the clarity of communication but also ensure that the expressions used by humans in sign language are accurately transposed to those who may not be fluent in it.

A particular challenge lies in enhancing the performance of sign language recognition systems when applied to continuous video streams. Unlike image-based recognition, where individual signs can be accurately identified, video-based recognition requires the system to handle dynamic and fluid sequences of signs, often leading to decreased accuracy. Addressing this challenge is crucial for developing more effective and reliable sign language recognition technologies, especially for real-time applications such as live communication and education. Overcoming these obstacles will be an important step toward offering an accessible experience for users who rely on sign language.

These advancements could play a pivotal role in bridging the communication gap for sign language users, ultimately contributing to a more inclusive and accessible environment. As the field continues to evolve, focusing on these innovative strategies will be essential for enabling better interaction and understanding among diverse communities.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
ML	Machine learning
WCAG	Content Accessibility Guidelines
TTS	Text-to-speech
SR	Speech recognition

SLR	Sign language recognition
STT	Speech-to-text
PWD	People with disabilities
CNN	Convolutional neural network
VAPAs	Voice-activated personal assistants
MOS	Mean opinion score
ASR	Automatic speech recognition
ISL	Indian sign language
ASL	American sign language
GLLM	Generative large language models

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